

An introduction to optimal control and controlled wave problem

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17/2/2022

BASIC KNOWLEDGE

Finite dimensional system

A finite dimensional control system has a form

$$x' = Ax + Bv, \quad 0 \leq t \leq T, \quad x(0) = x^0, \quad (1)$$

where x is N -dimensional state and v is M -dimensional control,
 $M \leq N$.

$A : N \times N$ matrix.

$B : N \times M$ matrix.

Controllability

A system (1) is said to be controllable in time T if and only if for every x_0 , there exists a control v such that $x(T) = x_T$ for any given $x_T \in \mathbb{R}^N$.

Null controllability

A system (1) is controllable if and only if this system is null-controllable (it is true since we consider linear finite dimensional system). Particularly, for every initial datum x_0 , there exists a control v such that $x(T) = 0$.

Kalman rank condition

A system (1) is controllable if and only if

$$\text{rank}[B, AB, \dots, A^{N-1}B] = N. \quad (2)$$

CONTINUOUS WAVE EQUATION

The control problem

Controlled wave equation

$$\begin{cases} y_{tt} - c^2 y_{xx} = 0, & 0 < x < 1, 0 < t < T, \\ y(0, t) = 0; y(1, t) = v(t), & 0 < t < T, \\ y(x, 0) = y^0(x), y_t(x, 0) = y^1(x), & 0 < x < 1, \end{cases} \quad (3)$$

where $(y^0(x), y^1(x)) \in L^2(0, 1) \times H^{-1}(0, 1)$.

The problem is to find the control $v(t)$ satisfy
 $y(x, T) = y_t(x, T) = 0$.

The formula of the control

J.L.Lions has indicated that the control $v(t)$ with minimum cost (norm) is $u_x(1, t)$, where the solution u satisfies the system

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0, & 0 < x < 1, 0 < t < T, \\ u(0, t) = u(1, t) = 0, & 0 < t < T, \\ u(x, 0) = u^0(x), u_t(x, 0) = u^1(x), & 0 < x < 1, \end{cases} \quad (4)$$

where $(u^0, u^1) \in H^1(0, 1) \times L^2(0, 1)$.

Minimization problem

The different model of the control problem is to minimize the functional J determined by

$$J((u^0, u^1)) = \frac{1}{2} \int_0^T |u_x(1, t)|^2 dt + \int_0^1 y^0 u^1 dx - \langle y^1, u^0 \rangle_{H^{-1} \times H^1} . \quad (5)$$

Observability inequality

The control problem of (3) is equivalent to the observability inequality. Particularly, find a constant $C(T) > 0$ such that

$$E(0) \leq C(T) \int_0^T |u_x(1, t)|^2 dt, \quad (6)$$

where u is the solution of

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0, & 0 < x < 1, 0 < t < T, \\ u(0, t) = u(1, t) = 0, & 0 < t < T, \\ u(x, 0) = u^0(x), u_t(x, 0) = u^1(x), & 0 < x < 1, \end{cases} \quad (7)$$

and the energy $E(0)$

$$E(0) = E(t) = \frac{1}{2} \int_0^1 [|u_x(x, t)|^2 + |u_t(x, t)|^2] dx. \quad (8)$$

Uniform observability estimate

Proposition

For an arbitrary $T \geq 2$, system (7) is observable. Otherwise, when $T < 2$, we have

$$\sup_u \left[\frac{E(0)}{\int_0^T |u_x(1, t)|^2 dt} \right] = \infty, \quad (9)$$

where $\sup$ takes over all solutions u of (7).

Idea: + Consider the Fourier series of solution u .

+ Represent the energy $E(0)$ and the integral $\int_0^T |u_x(1, t)|^2 dt$.

HUM method and operator Λ

Set $E = H_0^1(0, 1) \times L^2(0, 1)$ and $E' = H^{-1}(0, 1) \times L^2(0, 1)$.

Consider solution u satisfy (7) for any $e = \{u^0(x), u^1(x)\} \in E$.

Subsequently, compute $u_x(1, t)$ and solve the problem

$$\begin{cases} y_{tt} - c^2 y_{xx} = 0, & 0 < t < T, 0 < x < 1, \\ y(0, t) = 0, y(1, t) = u_x(1, t), & 0 < t < T, \\ y(x, T) = y_t(x, T) = 0, & 0 < x < 1. \end{cases} \quad (10)$$

Finally, Lions defines the operator $\Lambda : E \rightarrow E'$ by

$$\Lambda(e) = \{y_t(x, 0), -y(x, 0)\}. \quad (11)$$

HUM method and operator Λ

Given $f = \{y^1, -y^0\} \in E'$, HUM method is to solve the equation

$$\Lambda e = f. \quad (12)$$

After receiving the solution $e(x) = \{u^0(x), u^1(x)\}$, we solve the (7).

The $v(t) = u_x(1, t)$ is control that satisfy the null-controllability.

SEMI DISCRETE WAVE EQUATION

The control problem

Given an integer N , we define $h = \frac{1}{N+1}$. We divide the interval $(0, 1)$ into $N + 1$ intervals which is determined by $I_j = [x_j, x_{j+1}]$ $j = \overline{0, N+1}$ where $x_i = ih$. The control wave equation problem is

$$\begin{cases} y_j''(t) - c^2 \frac{1}{h^2} [y_{j+1}(t) - 2y_j(t) + y_{j-1}(t)] = 0, & 0 < t < T, j = \overline{1, N}, \\ y_0(t) = 0; y_{N+1}(t) = v(t), & 0 < t < T, \\ y_j(0) = y_j^0, y'(0) = y_j^1, & j = \overline{1, N}. \end{cases} \quad (13)$$

Non uniform observability estimate

In the semidiscrete case, the solution u in (5) and the energy $E(0)$ in (6) defined by

$$\begin{cases} u_j''(t) - c^2 \frac{1}{h^2} (u_{j+1}(t) - 2u_j(t) + u_{j-1}(t)) = 0, & 0 < t < T, j = \overline{1, N}, \\ u_0(t) = u_{N+1}(t) = 0, & 0 < t < T, \\ u_j(0) = u_j^0, u_j'(0) = u_j^1, & j = \overline{1, N}, \end{cases} \quad (14)$$

and

$$E_h(0) = E_h(t) = \frac{h}{2} \sum_{j=0}^N \left[(u_j'(t))^2 + \left(\frac{u_{j+1}(t) - u_j(t)}{h} \right)^2 \right]. \quad (15)$$

Non uniform observability estimate

Theorem

For any $T > 0$, when $h \rightarrow 0$, we have

$$\sup_u \frac{E_h(0)}{\int_0^T |u_N/h|^2 dt} \rightarrow \infty, \quad (16)$$

where u is solution of (14).

Idea: + Choose a specific solution u_h satisfy (14).

+ Compute $\frac{E_h(0)}{\int_0^T |u_N/h|^2 dt}$ specifically and take the limit when $h \rightarrow 0$.

Consequence

Proposition ([2])

The control $v_h(t) = \frac{u_N(t)}{h}$ associated with the semidiscrete system (14) will be divergent when $h \rightarrow 0$, i.e. $\|v_h\|_{L^2(0,T)}$ is not bounded.

Idea: + Prove by contradiction.

FULLY DISCRETE SCHEME

Fully discrete scheme

Given two integers M and N , we define mesh point $\{x_j\}_{j=0}^{N+1}$ and $\{t_k\}_{k=0}^{M+1}$. Set $\Delta x = \frac{1}{N+1}$ and $\Delta t = \frac{T}{M+1}$. System (4) can be discretized by

$$\begin{cases} \frac{u_j^{k+1} - 2u_j^k + u_j^{k-1}}{(\Delta t)^2} - c^2 \frac{u_{j+1}^k - 2u_j^k + u_{j-1}^k}{(\Delta x)^2} = 0, & k = \overline{1, M}, j = \overline{1, N}, \\ u_0^k = u_{N+1}^k = 0, & k = \overline{0, M+1}, \\ u_j^0 = e_j^0, \frac{u_j^1 - u_j^0}{\Delta t} = e_j^1, & j = \overline{0, N+1}, \end{cases} \quad (17)$$

Fully discrete scheme

System (3) is determined by

$$\begin{cases} \frac{y_j^{k+1} - 2y_j^k + y_j^{k-1}}{(\Delta t)^2} - c^2 \frac{y_{j+1}^k - 2y_j^k + y_{j-1}^k}{(\Delta x)^2} = 0, & k = \overline{1, M}, j = \overline{1, N}, \\ y_0^k = 0, y_{N+1}^k = -\frac{u_N^k}{\Delta x}, & k = \overline{0, M+1}, \\ y_j^{M+1} = 0, \frac{y_j^{M+1} - y_j^M}{\Delta t} = 0, & j = \overline{0, N+1}. \end{cases} \quad (18)$$

Nonuniform observability estimate

The energy $E(0)$ defined by (6) and $\int_0^T |u_x(1, t)|^2 dt$ are determined by

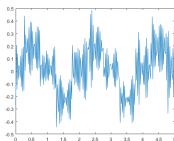
$$E_{\Delta x, \Delta t}^n = \frac{\Delta x}{2} \sum_{j=1}^N \left(\frac{u_j^{n+1} - u_j^n}{\Delta t} \right)^2 + \frac{\Delta x}{2} \sum_{j=1}^{N+1} \frac{u_j^{n+1} - u_{j-1}^{n+1}}{\Delta x} \frac{u_j^n - u_{j-1}^n}{\Delta x}, \quad (19)$$

and

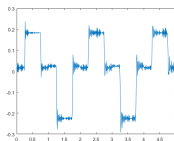
$$\|u_N / \Delta x\|_{l^2(\Delta t)}^2 = (\Delta t) \sum_{k=0}^{M+1} \left(\frac{u_N^k}{\Delta x} \right)^2. \quad (20)$$

Numerical result

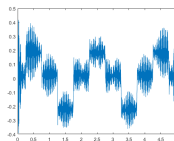
Numerical experiment with initial data y_0 piecewise constant function and $y_1 \equiv 0$.



(a) $N = 51$



(b) $N = 101$



(c) $N = 201$

CURE METHOD

Cures

There are several ways to cure the divergence of the control v proposed by R.Glowinski, C-H.Li and J.-L. Lions in [1].

- (a) Tikhonov regularization.
- (b) Bigrid method.
- (c) Fourier filtering.

Tikhonov regularization and Glowinski

In [1], in stead of solving (12), Glowinski solves the following equation

$$\begin{cases} \Lambda e_\epsilon + \epsilon M e_\epsilon = f, & x \in (0, 1) \\ e_\epsilon(x) = 0, \partial_{xx} e_\epsilon^0(x) = 0, & x \in \{0, 1\}, \end{cases} \quad (21)$$

where $e_\epsilon = \{e_\epsilon^0, e_\epsilon^1\}$ belongs to $H = H^2(0, 1) \cap H_0^1(0, 1) \times H_0^1(0, 1)$ and $M = \begin{pmatrix} \partial_{xxxx} & 0 \\ 0 & -\partial_{xx} \end{pmatrix}$.

Tikhonov regularization

We can discretize the equation (13) as follow.

$$\begin{cases} \Lambda_h e_h + h^2 M_h e_h = f, & x \in (0, 1), \\ e_{h,j} = 0, & j = 0; N + 1, \end{cases} \quad (22)$$

$$\text{where } M_h = \begin{pmatrix} \frac{1}{h^4} A & 0 \\ 0 & -\frac{1}{h^2} B \end{pmatrix}, \quad A = \begin{pmatrix} 5 & -4 & 1 & \dots & 0 \\ -4 & 6 & -4 & 1 & \dots \\ 1 & -4 & 6 & -4 & 1 \\ \dots & 1 & -4 & 6 & -4 \\ \dots & 0 & 1 & -4 & 5 \end{pmatrix}$$

$$\text{and } B = \begin{pmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots \\ \dots & 0 & 1 & -2 & 1 \\ \dots & 0 & 0 & 1 & -2 \end{pmatrix}.$$

In [1], Glowinski said that $\epsilon = h^2$ is efficient to solve (22).

Fourier filtering

Consider the space

$$C_\gamma(h) = \left\{ u : u = \sum_{k=1}^{[\gamma/h]} \left(a_k \cos(\sqrt{\lambda_k^h} t) + \frac{b_k}{\sqrt{\lambda_k^h}} \sin(\sqrt{\lambda_k^h} t) \right) w_k^h \right\}, \quad (23)$$

where $\lambda_k^h = \frac{4}{h^2} \sin^2 \left(\frac{k\pi h}{2} \right)$ and

$w_k^h = (w_{k,1}, \dots, w_{k,N})^T$, $w_{k,j} = \sin(k\pi jh)$, $j = 1, \dots, N$.

The Fourier filtering is to solve the system (4) over the space

$C_\gamma(h)$.

Fourier filtering

Theorem ([2])

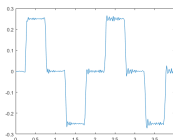
Given an arbitrary $\gamma > 0$, there exists time $T(\gamma) > 0$ such that for all $T > T(\gamma)$, there exists $C(T, \gamma) > 0$ such that

$$E_h(0) \leq C(T, \gamma) \int_0^T \left| \frac{u_N(t)}{h} \right|^2 dt, \quad (24)$$

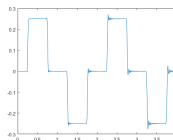
for every solution u of system (4) in the class $C_\gamma(h)$ and for all $h > 0$. Furthermore, if we have $T(\gamma) \rightarrow 2$ as $\gamma \rightarrow 0$ and $T(\gamma) \rightarrow \infty$ when $\gamma \rightarrow 1$.

Numerical result, Tikhonov

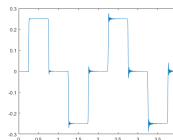
Numerical experiment with initial data y_0 piecewise constant function and $y_1 \equiv 0$.



(a) $N = 51$



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(c) $N = 201$

Reference



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